Draft proof for ACI with delay

Lemma 1 (Lemma 4.1 [1]): With probability one we have that $\forall t \in \mathbb{N}, \alpha_t \in [-\gamma, 1+\gamma]$.

Recall that the update formula for α_t is

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t). \tag{1}$$

Proof: Assume by contradiction that with positive probability $\{\alpha_t\}_{t\in\mathbb{N}}$ is such that $\inf_t \alpha_t < -\gamma$ (the case where $\sup_t \alpha_t > 1+\gamma$ is identical). Note that $\sup_t |\alpha_{t+1} - \alpha_t| = \sup_t \gamma|\alpha - \text{err}_t| < \gamma$. Thus, with positive probability we may find $t \in \mathbb{N}$ such that $\alpha_t < 0$ and $\alpha_{t+1} < \alpha_t$. However,

$$\begin{aligned} \alpha_t < 0 \implies \hat{Q}_t(1-\alpha_t) = \infty \implies \text{err}_t = 0 \\ \implies \alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t) \geq \alpha_t \end{aligned}$$

and thus $\mathbb{P}(\exists t \text{ such that } \alpha_{t+1} < \alpha_t < 0) = 0$. We have reached a contradiction. ∎

Lemma 2: With probability one we have that $\forall t \in \mathbb{N}, \alpha_t \in [-\gamma k(1-\alpha), 1+\gamma k\alpha]$.

We could scale the bounds to $[-\gamma k, 1+\gamma k]$ for simplicity.

Proof: We first rewrite the update formula for α_t as

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_{t-k+1}). \tag{2}$$

Assume by contradiction that with positive probability $\{\alpha_t\}_{t\in\mathbb{N}}$ is such that $\inf_t \alpha_t < -k\gamma(1-\alpha)$ (the case where $\sup_t \alpha_t > 1+k\gamma\alpha$ is identical).

The lower bound (α_t is decreasing): note that

$$\begin{aligned} \alpha_{t+1} - \alpha_{t-k+1} &= \sum_{j=t-k+1}^{t} \gamma(\alpha - \text{err}_{j-k+1}) \quad \text{(by Eq. (2))} \\ &\geq -k\gamma(1-\alpha). \end{aligned}$$

(since α is small and $\alpha - \text{err}$ is at its maximum when $\text{err} = 1$)

Thus, we have

$$\alpha_{t+1} \geq \alpha_{t-k+1} - k\gamma(1-\alpha) \geq -k\gamma(1-\alpha),$$

which contradict with $\alpha_{t+1} < -\gamma k(1-\alpha)$.

The upper bound (α_t is increasing): note that

$$\begin{aligned} \alpha_{t+1} - \alpha_{t-k+1} &= \sum_{j=t-k+1}^{t} \gamma(\alpha - \text{err}_{j-k+1}) \quad \text{(by Eq. (2))} \\ &\leq k\gamma\alpha. \end{aligned}$$

(since α is small and $\alpha - \text{err}$ is at its maximum when $\text{err} = 0$)

Thus, we have

$$\alpha_{t+1} \leq \alpha_{t-k+1} + k\gamma\alpha \leq 1 + k\gamma\alpha,$$

which contradict with $\alpha_{t+1} > 1 + k\gamma\alpha$. ∎

Proposition 1 (Proposition 4.1 [1]): With probability one we have that for all $T \in \mathbb{N}$,

$$\left| \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t - \alpha \right| \leq \frac{\max\{\alpha_1, 1 - \alpha_1\} + \gamma}{T\gamma}. \tag{3}$$

In particular, $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t \stackrel{a.s.}{=} \alpha$.

Proof: By expanding the recursion defined in Eq (1) and applying Lemma 1 we find that

$$[-\gamma, 1+\gamma] \ni \alpha_{T+1} = \alpha_1 + \sum_{t=1}^{T} \gamma(\alpha - \mathrm{err}_t).$$

Rearranging this gives the result. ■

Proposition 2 (Proposition 4.1 [1]): With probability one we have that for all $T \in \mathbb{N}$,

$$\left| \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t - \alpha \right| \leq \frac{\max\{\alpha_k, 1 - \alpha_k\} + k\gamma}{T\gamma}.$$

In particular, $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t \stackrel{a.s.}{=} \alpha$.

Proof: By Eq. (2), we have

$$\sum_{t=k}^{T+k-1} (\alpha_{t+1} - \alpha_t) = \sum_{t=k}^{T+k-1} \gamma(\alpha - \mathrm{err}_{t-k+1}).$$

Specifically, the left hand side (LHS) is a telescoping sum, and we have

$$\begin{aligned} \mathrm{LHS} = (\alpha_{T+k} - \alpha_{T+k-1}) + (\alpha_{T+k-1} - \alpha_{T+k-2}) + \\ \cdots + (\alpha_{k+1} - \alpha_k) \\ = \alpha_{T+k} - \alpha_k. \end{aligned}$$

For the right hand side (RHS), let $j = t - k + 1$, we have

$$\mathrm{RHS} = \gamma \sum_{j=1}^{T} (\alpha - err_j),$$

leading

$$\alpha_{T+k} - \alpha_k = \gamma(T\alpha - \sum_{j=1}^{T} err_j) \implies$$

$$\frac{1}{T} \sum_{j=1}^{T} err_j - \alpha = \frac{\alpha_k - \alpha_{T+k}}{T\gamma} \implies$$

(By dividing both sides by $T\gamma$ and rearrange the terms)

$$\left| \frac{1}{T} \sum_{j=1}^{T} err_j - \alpha \right| = \frac{|\alpha_k - \alpha_{T+k}|}{T\gamma}$$

(By taking absolute value)

Furthermore, we derive that

$$\begin{aligned} |\alpha_k - \alpha_{T+k}| &\leq \max\{(1 - \alpha_k) + k\gamma\alpha, \alpha_k + k\gamma(1 - \alpha)\} && \text{(By Lemma 2)} \\ &\leq \max\{(1 - \alpha_k) + k\gamma, \alpha_k + k\gamma\} && \text{(Since } \alpha \in [0,1]) \\ &= \max\{(1 - \alpha_k), \alpha_k\} + k\gamma \end{aligned}$$

■

References

[1] I. Gibbs and E. Candes, "Adaptive conformal inference under distribution shift," *Advances in Neural Information Processing Systems*, vol. 34, pp. 1660–1672, 2021.